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Pakistan Studies
Final assignment



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ENERGY CRISIS IN PAKISTAN (ALTERNATE SOURCES)

Introduction:

The energy crisis stems from the foreseeable end of the cycle of oil, gas and coal, which, in addition, have been producing a considerable increase in greenhouse gases.

In recent years, many scientists have raised their voice to warn about climate change, caused notably by the burning of oil and coal in order to produce energy.

Global energy consumption is increasing and we will face a shortage of fossil fuels in the coming decades. Therefore, the availability of reserves is an important source of concern.

Our current consumption model relies almost entirely on the use of non-renewable energy sources such as oil, gas, coal and uranium. At the current rate of consumption, oil will be the first fossil fuel to run out. According to projections, there would be between 40 and 60 years of proven reserves of conventional oil. Natural gas could be exploited for another 70 years. For coal, there would be around two centuries of reserves.

These data are to be put into perspective because they are based on current consumption, while it is clear that it will increase considerably. Energy demands are and will be amplified by the demographic - the world's population should reach nearly 10 billion people in 2050 - and economic boom of growing areas. According to the International Energy Agency (IEA), global energy demand could increase by more than 50% by 2030 in the absence of public policies in this area.

Another reason for energy shortage and scarcity is the poor infrastructure of power generating equipment. Most of energy producing companies keep on using outdated equipments that limits energy production. The need to upgrade the infrastructure and set a high standard of performance is critical.

Mainly coming from the unnecessary use of energy resources, energy waste describes the wastage of energy sources, in particular fuels and electricity. Consequently, the

reduction of waste is a colossal source of energy savings, which requires actions both on an individual and collective level.

Energy crisis in Pakistan:

According to the Ministry of Finance, the energy crisis is the largest single drain on Pakistan's economy, shaving off up to 2 percentage points from annual gross domestic product growth in the country (see EAW, 2013: "Overview"). The current energy crisis began to manifest itself in earnest by late 2007. Although the causes of the crisis are structural, the immediate trigger was the 2007 global commodity price boom, when oil prices almost tripled over an 18-month period.¹ The unprecedented fuel inflation was a key factor in the 36% increase in Pakistan's import bill in fiscal year 2008 (see EAW, 2013: "Statistical appendix", Table 8.3), and the consequences of an energy generation policy that relies heavily on oil-fired thermal generation became all too clear.

This expert analysis briefly discusses the key issues affecting the energy sector, and then analyses how energy issues are having an impact on growth and development in Pakistan. It focuses particularly on the new energy policy issued by the government that came to power in May 2013.

The following sections delineate the origins of the energy crisis and some of the key issues that the government is currently grappling with. The expert analysis then goes on to look at the solutions that the government is trying to implement and assesses whether these will indeed prove to be effective in reducing energy shortages and reviving growth.

Current issues:

The key fiscal issues around the energy crisis currently being experienced by the Pakistani economy are discussed below.

Fuel mix:

cost implications The fuel mix transformation described earlier has cost implications, because power generation from imported furnace oil is significantly more expensive than from hydro sources – approximately Rs. 12-17 per unit of oil power generation, compared to hydro generation, which costs about Rs. 1 per unit (see Pakistan, 2013).³ Generation from diesel-fired power plants is even more expensive at about Rs. 23 per unit of power generated.

Circular debt:

The increased dependence on expensive, thermal oil power generation has also given rise to the phenomenon of circular debt in the energy sector, in terms of which slippages in bill payments (particularly on the part of public institutions) trigger off a chain of delayed payments for imported furnace oil, natural gas or other inputs to the thermal generation system, which in turn hamper the operation of the power plants and result in less than optimum capacity usage. As of the end of July 2013 the circular debt had increased to almost \$5 billion, and was being cited as a significant drag on the power sector and a key factor impeding the efficient operation of independent power producers.

POLICY IMPLICATIONS In Pakistan:

- Pakistan should consolidate its many energy-related institutions into a single ministry. This will bring some urgently needed order and efficiency to its dysfunctional energy sector.
- A short-term fix that could bring immediate relief is to request a new loan from the International Monetary Fund (IMF). However, because the IMF would probably impose politically delicate conditions, Islamabad is unlikely to make such a request until after this spring's elections.
- Tax reform is imperative and should be designed to provide Islamabad with more revenue to address the energy crisis.
- Pakistan can initially better diversify its energy mix by importing clean coal, which is often cheaper than imported oil and gas.
- Pakistan will not be able to implement the reforms needed to resolve its energy crisis unless Pakistanis elect leaders this spring who genuinely desire to serve the interests of their country.

A Lack of Strategy and Political Will:

A subset of the energy financing problem is an inability or unwillingness to muster the necessary political will to address the money shortage. More broadly, Pakistan has never developed a comprehensive, integrated energy strategy, and Islamabad's haphazard policies have failed to address the crisis's deep roots. The problem lies not with civil servants, bureaucrats, and technical experts who focus on developing

energy policies (many of them reasonable and actionable), but rather with the non-expert, high-level political appointees spread across the energy sector and beyond who are charged with implementing them.

Pakistan announced a national energy plan in 2010, though it was dominated by much-mocked—and likely ignored—conservation measures, such as bans on all-night wedding parties and neon billboards, along with the required early closures of street markets. (A more realistic demand-management strategy, announced last year by the Asian Development Bank, calls for the distribution of twenty million low-energy light bulbs.) Other well-intentioned initiatives have likewise not produced results. Pakistan has established the National Electric Power Regulatory Authority (NEPRA), charged with ensuring fair energy competition and consumer protection, but political interference undermines its autonomy and effectiveness. Tariff decisions must be approved by Islamabad, and NEPRA's four members are all selected by the government. Furthermore, government officials have been known to outright ignore the body's decisions.

Recent recommendations put forth by Pakistan's Planning Commission, however, offer some hope. In 2011, the commission released what it described as a “new framework” for economic growth, which calls for more focus on the private sector, cities, and youth. If implemented, the impact would be immense, as the plan would represent a paradigm shift in Pakistan's development philosophy.

In the context of energy, the document proposes some of the most far-reaching and comprehensive policy measures ever introduced in Pakistan—from full-scale sectoral deregulation to governance reform and the phasing out of many subsidies. Unfortunately, there are several problems. One is that while the Planning Commission is part of the government, it lacks implementation power, and no government entity has stepped up to embrace the commission's ideas and take on the mantle of implementation. (In fact, government agencies often spar with the Planning Commission.) Another dilemma is that the Planning Commission insists that such measures are only implementable after the country has established an integrated energy policy, which has still not happened. Moreover, Islamabad likely has little desire to authorize the Planning Commission's measures anyway, given that some of them (such as phasing out subsidies) are fraught with political risk—especially with national elections scheduled for spring 2013. On the subsidy question, in particular, while many experts accurately note that Islamabad's policies distort pricing, these measures are unlikely to change for political (and not systemic) reasons. This is because subsidies provide temporary relief to an impoverished mass population that often harbors antigovernment sentiment. It also bears mentioning that reducing subsidies could have an unintended effect:

increasing the number of Pakistanis who do not pay their taxes (given that if the poor are asked to pay more for energy, they may not have enough money to pay their taxes).

Yet herein lies a major dilemma, because Pakistan's government would significantly increase its revenue—and hence its ability to pay its energy bills—if more of the country paid its taxes. Former U.S. secretary of state Hillary Clinton has claimed that only 2 million of Pakistan's population of 180 million pay income taxes, while Pakistan's Federal Board of Revenue estimates that 700,000 wealthy Pakistanis are not paying their returns. The latter figure, in particular, suggests that revenue gains from increasing the number of citizens paying taxes could be tremendous. However, the government refuses to pressure its most affluent citizens, because many of them are politically connected or politicians themselves. And admittedly, there is no guarantee that Islamabad would actually use this added tax revenue to cover its energy debt; it could well spend the revenue on the repayment of other debts, administrative costs, or even defense.

Scenarios for the Future:

Given that Pakistan lacks the revenue to finance an energy recovery, future opportunities abound for international donors, including the United States. Washington, in fact, already provides a considerable amount of energy assistance to Pakistan. The Obama administration identifies energy as a priority area in its civilian assistance program to the country, and Congress released nearly \$300 million in new energy aid last summer alone. The U.S. private sector also contributes to Pakistan's energy sector, including in overtly political ways. Last summer, ConocoPhillips mediated talks between Pakistan and Qatar on a potential natural gas deal in an effort to discourage Islamabad from pursuing a pipeline project with Iran that is opposed by Washington.

Given the extent of Pakistan's energy woes, and especially its circular debt—which, at its highest point of nearly \$4.5 billion, far exceeded the value of Washington's \$1.5 billion in total annual civil assistance—it is folly to expect U.S. energy aid to make a major dent in the crisis. Conversely, if U.S. civilian assistance to Pakistan were to be cut, the reduction of energy-intensive aid would be a significant loss for the country. This assistance includes a USAID infrastructure project, expected to be completed by this year, which adds nearly 1,000 MW to the national power grid—a fifth of Pakistan's energy shortfall.

This is not to say that indigenous solutions should simply be discarded. Consider the vast Thar coalfields in Sindh Province, where 200 billion tons of reserves have

lay dormant since their discovery more than twenty years ago (Thar constitutes the world's sixth-largest coal deposit). Last year, Islamabad designated Thar as a special economic zone, hoping to lure investors with tax breaks and other incentives. Some, however, believe that the government must be more aggressive. Thar has been a common theme on the campaign trail for this year's elections, with opposition parties hailing Thar as an elixir for Pakistan's energy supply crunch and underscoring the urgency of tapping into its riches.

However, what both the government and political opposition fail to articulate is how Pakistan will overcome the formidable challenge of developing the technological and labor capacity to exploit this potential bonanza. Another problem is purely political. Ever since the Thar coal was discovered, the central government has been locked in a disagreement with the Sindh provincial government about how to divvy up the spoils. Islamabad has proposed an 80/20 split, while Sindh has insisted that it retain full control of the coalfields. This 22-year-old disagreement has effectively put on hold the exploitation of Thar's resource treasures and crystallizes how Pakistan's energy woes are as much (if not more) a governance and political issue as one of supply and demand.

Encouragingly, Pakistan is also starting to explore other alternative energy sources. Officials have said several small-scale wind projects are under construction. The government has also announced that by 2030 it plans to have a minimum of 5.0% of total commercial energy supply provided by wind, solar, and biowaste, and that 2.5% of Pakistan's overall energy generation will come from renewables. Islamabad claims that by 2030 about 5,500 MW of Pakistan's projected 160,000-MW daily energy requirement will come from alternative and renewable sources. These are admittedly ambitious goals, given the miniscule role renewables play in the current energy mix.

Ultimately, it is the issue of implementation that prolongs Pakistan's energy crisis, making many experts pessimistic that the crisis can be resolved anytime soon. There is no shortage of research, conferences, and proposals offering policy solutions. However, these measures are not executed, because there is no political will to do so. This has long been the case at both federal and provincial levels, as well as with different political parties. While the ruling Pakistan Peoples Party has been the political face of the energy crisis since 2008, the previous government (led by Pervez Musharraf and the Pakistan Muslim League Quaid-e-Azam, or PML-Q) largely restricted its energy policies to supply-generation measures—the same politically safe bets made today. Then, as now, few efforts were made to strengthen energy governance or reform the energy pricing system. Tellingly, even in the rare cases when the government enacts politically risky measures to

strengthen the energy sector and overall economy, it often reverses course. In 2011, for example, Islamabad repealed an increase in fuel prices—instituted to raise desperately needed revenue—after a key coalition partner had withdrawn from the government to protest this price hike.

Implications for growth:

The three issues delineated above do not by any means constitute a comprehensive picture of the problems that beset the energy sector in Pakistan. In particular, the technical and operational problems that plague the transmission and distribution system, fuel adulteration issues, supply chain blockages that prevent efficient generation, and the poor governance that enables the widespread theft of electricity and the non-payment of bills are issues that require more detailed analysis than it is possible to incorporate in this discussion.

Nevertheless, the three issues described go a long way in explaining how Pakistan's economy has been crippled by the energy crisis. To summarise, the fuel mix transformation of the mid-1990s effectively rendered the generation of power about ten times more expensive than it had been when hydropower was the key form of generation. The increase in generation costs precipitated increases in tariffs, which in turn triggered a round of payment defaults and delays on the part of a significant number of mainly

public sector entities, in addition to households. These instances of non-payment of dues have reverberated right through the supply chain, causing IPPs to operate at less than optimum capacity, thus precipitating a power crisis. With power shutdowns of 12-18 hours a day during summer months in major cities and smaller industrial towns, production processes have been badly disrupted and/or production costs increased as manufacturing enterprises and service outlets either reduce shifts and hours of operation or invest in expensive alternate power supply systems.

The disruption of power supplies is one side of the picture. In addition, the energy crisis is a significant drain on the government's resources, with energy subsidies taking up a substantial portion of the federal budget. The government's attempts to bail out the leading power sector utilities and ensure minimal return on equity to distribution companies is bleeding the exchequer and nullifying any reform initiatives that these entities may have in mind. Power sector utilities suffer from the "too big to fail" syndrome in terms of which they are justified in believing that the government will always step in to pay their deficits because they are providing

an essential resource. While it is true that the sector cannot be allowed to fail, the federal government cannot delay the reform of these entities and has to allow them to take the fall for decisions that lead to technical and financial losses.

The IMF's recommendations for the energy sector:

The energy crisis has become one of the key impediments to growth in Pakistan and as such has prompted international financial institutions to renew their interest in the energy sector. In terms of policy reform, the key document that will guide Pakistan's policy agenda in the short to medium term is the government's agreement with the International Monetary Fund (IMF) reached in September 2013, according to which a \$6.7 billion loan has been extended to Pakistan under the IMF's Extended Fund Facility. The agreement includes a number of clauses that specifically deal with the energy sector. In the document, IMF staff argued that the key to reform is to make the energy sector financially viable (see IMF, 2013: para. 38). The measures suggested to do this included clearance of the circular debt, tariff adjustments aimed at improving resource allocation and encouraging conservation, and fuel policies aimed at ensuring the supply of natural gas to power plants to facilitate a switch from furnace oil to a cheaper indigenous fuel.

Conclusion:

Pakistan's energy sector has become a major drain on the economy and is impeding growth, both because of power shortages (which have affected small manufacturing enterprises and services in particular) and because of the budgetary impacts of energy subsidies, which divert much-needed resources from more productive sectors.

The energy sector has become a focus of public policy in recent years and has garnered the attention of international financial institutions, including the IMF. Pressure for reform of the sector as a whole, and the power generation, transmission and distribution regime in particular, has grown substantially. The government is obliged to carry out tariff adjustments, remove subsidies and ensure a level playing field for all private sector entities active in power generation, in addition to other policy and governance reforms. The process has got off to a rocky start, with the judiciary calling into question the power sector regulator's authority and competence. How the government handles this delicate situation in the face of scepticism from state organs and hostility from consumers remains to be seen.